

Probability & Statistics

APM_2F010_EP

Lecture Notes

University: Ecole Polytechnique

Lecturer: Prof. Tardy Yoan

Student: Nikolay Kharovskiy

Semester: semester 3

Date: September 21, 2026

These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Lecture 1

Theorem 1: Fubini positive

Let $f : X \times Y \rightarrow [0, \infty)$, $g : X \rightarrow [0, \infty)$ and $h : Y \rightarrow [0, \infty)$ be piecewise continuous. Then:

$$\int_Y \left(\int_X f(x, y) g(x) dx \right) h(y) dy = \int_X \left(\int_Y f(x, y) h(y) dy \right) g(x) dx$$

- Works without integrability assumptions.
- Analogue: reordering double series with nonnegative terms.

Theorem 2: Fubini - Tonelli

Let $f : X \times Y \rightarrow \mathbb{R}$ (not necessarily positive), and let $g : X \rightarrow [0, \infty)$ and $h : Y \rightarrow [0, \infty)$ be piecewise continuous. If:

$$\int_Y \left(\int_X |f(x, y)| g(x) dx \right) h(y) dy < \infty$$

Then:

$$\int_Y \left(\int_X f(x, y) g(x) dx \right) h(y) dy = \int_X \left(\int_Y f(x, y) h(y) dy \right) g(x) dx$$

- Key assumption: integrable.

What is a probability?

Example 3: Rolling a fair dice

- Possible outcomes $\Omega = \{1, \dots, 6\}$.
- All outcomes have probability $1/6$.
- An event is “The result is odd”.

Proposition 4: Discrete probability space

$(\Omega, \mathbb{P})$ is a discrete prob. space. $\mathbb{P}$ is called the discrete prob. on Ω . If $\forall A \subset \Omega$, $\mathbb{P}(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}$, then $\mathbb{P}$ is called the uniform discrete probability.

Example 5: Non-discrete probability space

Let $E =$ “the extraction of a number in $[0, 1]$ uniformly at random”. E is modeled with the uniform probability $\mathbb{P}$ on the space $\Omega = [0, 1]$.

$$\forall 0 \leq a \leq b \leq 1, \quad \mathbb{P}([a, b]) = b - a$$

$(\Omega, \mathbb{P})$ cannot be a discrete prob. space.

Definition 6: Probability function

We define a prob. function $\mathbb{P} : \mathcal{P}([0, 1]) \rightarrow [0, 1]$ s.t.

1. $\mathbb{P}([0, 1]) = 1$
2. $\mathbb{P}$ is σ -additive
3. $\mathbb{P}([a, b]) = b - a \quad \forall a, b$